NNN-T-785B
July 15, 1969
SUPERSEDING
Fed. Spec. NNN-T-785A
January 31, 1963

FEDERAL SPECIFICATION

TUBE, CENTRIFUGE (PLASTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ungraduated, round bottom, plastic centrifuge tubes.

1.2 Classification. Plastic centrifuge tubes shall be of the following types and capacities, as specified (see 6.2):

Type I - Cellulose acetate butyrate.
- 10 - milliliter capacity.
- 20 - milliliter capacity.
- 50 - milliliter capacity.

Type II - Polypropylene.
- 12 - milliliter capacity.
- 18 - milliliter capacity.
- 50 - milliliter capacity.

Type III - Polycarbonate.
- 12 - milliliter capacity.
- 50 - milliliter capacity.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-349 - Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate.
L-P-393 - Plastic Molding, Material, Polycarbonate, Injection and Extrusion.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Setup.
PPP-C-843 - Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. 406 - Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington,D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-P-116 - Preservation, Methods of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Freight Classification

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Type I tubes. Type I centrifuge tubes shall be made from cellulose acetate butyrate conforming to type III of L-P-349.

3.1.2 Type II tubes. Type II centrifuge tubes shall be made from polypropylene.

TABLE I. Polypropylene properties

Requirement	Minimum	Maximum	Test method (Fed. Test Std. No. 406)
Specific gravity	0.897	0.903	5011
Tensile strength, p.s.i.	4900	-----	1011
Water absorption, percent	-----	0.01	7031
Brittleness, temperature, °C	-----	-15	2051

3.1.3 Type III tubes. Type III centrifuge tubes shall be made from polycarbonate resin conforming to L-P-393.

3.2 Style. All centrifuge tubes shall conform to the applicable capacities and dimensions of table II.

TABLE II. Capacities and dimensions of centrifuge tubes

Type	Capacity (ml)	Length (mm)	Outside diameter (mm)	Wall thickness (mm)
I	10	99 ± 2	13 ± 0.5	0.30 ± 0.08
I	20	114 ± 2	16 ± 0.5	0.30 ± 0.08
I	50	105 ± 2	29 ± 0.5	0.30 ± 0.08
II	12	100 ± 2	16 ± 1.0	1.0 ± 0.1
II	18	115 ± 2	19 ± 1.0	1.0 ± 0.1
II	50	105 ± 2	28 ± 1.0	1.0 ± 0.1
III	12	99 ± 2	16 ± 1.0	1.1 ± 0.15
III	50	104 ± 2	29 ± 1.0	1.3 ± 0.2

3.2.1 Type I tubes. Type I centrifuge tubes shall be ungraduated and transparent and shall have straight walls and round bottoms.

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3.2.2 Type II tubes. Type II centrifuge tubes shall be ungraduated and translucent and shall have straight walls and round bottoms.

3.2.3 Type III tubes. Type III centrifuge tubes shall be ungraduated and shall have a glass like clarity. The tubes shall be straight walls, round bottoms, and mouth openings having no lip or spout.

3.3 Resistance to centrifugal force. The tubes shall not break or crack when tested as specified in 4.2.4.1.

3.4 Resistance to heat (type III tubes only). Type III tubes shall not soften, deform, crack or become opaque when tested as specified in 4.2.4.2.

3.5 Workmanship. The tubes shall be free of striae, thin spots, bubbles, dents, cracks, and other defects which may detract from their appearance or impair their serviceability.

3.6 Manufacturer identification. Each centrifuge tube shall be permanently and legibly marked with the name or trademark of the manufacturer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of plastic centrifuge tubes of the same type and capacity produced by one manufacturer and offered for acceptance at one time.

4.2.2 Sampling.

4.2.2.1 For examination.

(a) Centrifuge tubes. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.
(b) Preparation for delivery. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-4.

4.2.3 Inspection procedure.

4.2.3.1 For examination.

(a) Centrifuge tubes. Sampling tubes taken in 4.2.2.1 (a) shall be examined for the following defects using an acceptable quality level (AQL) of 2.5 percent defective:

Categories	Defects
Critical:	None defined
Major:	
101	Dimension (length, outside diameter, or wall thickness)
102	Shape (side walls not straight or bottom not round)
103	Tube not ungraduated
104	Type I tubes not transparent
105	Type II tubes not translucent
106	Type III tubes not of glasslike clarity
107	Type III tubes not without lip or spout
108	Manufacturer identification (name or trademark) missing
109	Workmanship (see 3.5).

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(b) Preparation for delivery. The sample unit shall be one unit, intermediate, or shipping container, as applicable, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined for the following defects using an AQL of 4.0 percent defective:

Categories	Defects
Critical:	None defined
Major:	
101	Contents per container not as specified
102	Container not as specified
103	Container closure not as specified
104	Container damaged
105	Cushioning material incorrect or missing
106	Marking incorrect, missing, or illegible

4.2.3.2 For test. Sample tubes taken in 4.2.2.2 shall be tested as specified in the applicable portions of 4.2.4 using an AQL of 1.0 percent defective.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Resistance to centrifugal force.

(a) Apparatus. Any type or model centrifuge commercially available or normally employed in chemical laboratories may be used. The centrifuge shall be capable of speeds over 25,000 revolutions per minute (r.p.m.). The head of the centrifuge shall be such that the tips of the opposite centrifuge tubes are 8 to 9 inches apart when the equipment is operating. If a centrifuge with an 8 to 9 inch swing is not available, a machine with larger or smaller diameter swings may be used. However, the required centrifuge speed shall be determined as follows:

$$\text{Required r.p.m.} = 265\sqrt{RCF/D}$$

where: RCF = Relative centifugal force of 74,500 and
D = Diameter of the swing, in inches (measured from tip to tip of opposing centrifuge tubes).

(b) Procedure. The sample centrifuge tubes shall be tested in pairs. Fill each tube with tap water equal in quantity to the capacity of the tube being tested (see table I). Place the filled tubes in the centrifuge in balanced positions and centrifuge at 25,000 r.p.m. (or the r.p.m. determined for larger or smaller apparatus) for 30 minutes. The centrifuge shall be turned off and allowed to stop without use of the hand brake. Empty and examine the tubes. There shall be no evidence of cracks or breakage. Tested tubes shall be destroyed or reprocessed.

4.2.4.2 Resistance to heat (type III tubes only). Place the tube in an autoclave for no less than 30 minutes at a pressure of 15 pounds per square inch and a temperature of 121°C.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. One hundred centrifuge tubes of the same type and capacity shall be packaged in accordance with method III of MIL-P-116 in a type CF, class weather-resistant, variety SW, grade 275 fiberboard box of PPP-B-636. Cushioning material shall conform to type II, class B of PPP-C-843. The box shall be closed in accordance with method II given in the appendix of PPP-B-636.

5.1.1.2 Intermediate packaging. Four unit packages prepared as specified in 5.1.1.1 and containing centrifuge tubes of the same type shall be intermediately packaged in a type CF, class weather-resistant, grade V3c fiberboard box of PPP-B-636. The box shall be closed and sealed in accordance with the requirements of waterproofing of slotted style boxes.

5.1.2 Level C. Centrifuge tubes shall be packaged as specified in 5.1.1.1, except that the container shall conform to PPP-B-566 or PPP-B-676, type, variety, class, and style optional. Boxes shall be closed in accordance with the appendix of the applicable box specification.

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5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2). Shipping containers shall be of uniform shape and size, of minimum cube and tare and contain identical quantities of like items.

5.2.1 Level A. Two intermediate packages of 5.1.1.2 shall be packed in a class 2, style optional box of PPP-B-621 or an overseas type, style optional box of PPP-B-601. The boxes shall be closed and sealed in accordance with the appendix of the applicable box specification.

5.2.2 Level B. Two intermediate packages of 5.1.1.2 shall be packed in fiberboard boxes conforming to type CF, class weather-resistant, variety SW, grade 275 of PPP-B-636. The box shall be closed and sealed in accordance with method II given in the appendix of PPP-B-636.

5.2.3 Level C. Four packages of centrifuge tubes, packaged as specified in 5.1.2, shall be packed in a fiberboard box conforming to style RSC, type CF, class domestic, variety optional, grade 200 of PPP-B-636. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2), unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129 for military activities or Fed. Std. No. 123 for civil agencies.

6. NOTES

6.1 Intended use. Plastic tubes covered by this specification are intended for laboratory use in centrifuge equipment. They may also be used as test tubes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and capacity of centrifuge tubes (see 1.2).
(c) Level of packaging and packing desired (see 5.1 and 5.2).
(d) Additional marking, if required (see 5.3).

MILITARY INTEREST:

Custodians:

Army - MU
Navy - MS

Preparing activity:

Army - MU (EA)

Review activities:

Army - MD
Navy - MS
DSA - DP

User activities:

Army - MI, SM

CIVIL AGENCIES INTEREST:

AGR
COM - NBS
GSA - FSS
HEW
VA - DMS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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